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Clinical Pharmacokinetics of Paracetamol

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Summary

In therapeutic doses paracetamol is a safe analgesic, but in overdosage it can cause severe hepatic necrosis. Following oral administration it is rapidly absorbed from the gastrointestinal tract, its systemic bioavailability being dose-dependent and ranging from 70 to 90%. Its rate of oral absorption is predominantly dependent on the rate of gastric emptying, being delayed by food, propantheline, pethidine and diamorphine and enhanced by metoclopramide. Paracetamol is also well absorbed from the rectum. It distributes rapidly and evenly throughout most tissues and fluids and has a volume of distribution of approximately 0.9L/kg. 10 to 20% of the drug is bound to red blood cells.

Paracetamol is extensively metabolised (predominantly in the liver), the major metabolites being the sulphate and glucuronide conjugates. A minor fraction of drug is converted to a highly reactive alkylating metabolite which is inactivated with reduced glutathione and excreted in the urine as cysteine and mercapturic acid conjugates. Large doses of paracetamol (overdoses) cause acute hepatic necrosis as a result of depletion of glutathione and of binding of the excess reactive metabolite to vital cell constituents. This damage can be prevented by the early administration of sulfhydryl compounds such as methionine and N-acetylcysteine.

In healthy subjects 85 to 95% of a therapeutic dose is excreted in the urine within 24 hours with about 4, 55, 30, 4 and 4% appearing as unchanged paracetamol and its glucuronide, sulphate, mercapturic acid and cysteine conjugates, respectively. The plasma half-life in such subjects ranges from 1.9 to 2.5 hours and the total body clearance from 4.5 to 5.5ml/kg/min. Age has little effect on the plasma half-life, which is shortened in patients taking anticonvulsants. The plasma half-life is usually normal in patients with mild chronic liver disease, but is prolonged in those with decompensated liver disease.

Paracetamol (acetaminophen, N-acetyl-para-aminophenol, 4-hydroxy-acetanilide) was first used in medical therapy by von Mering in 1893. It has a molecular weight of 151.2 and is a moderately water- and lipid-soluble weak organic acid with a pKa of 9.5, and is thus largely un-ionised over the physiological range of pH. Although it has antipyretic and analgesic actions similar to aspirin (Beaver, 1966), it has only a weak anti-inflammatory action. In therapeutic doses it is a very safe analgesic, but in overdosage may cause severe hepatic necrosis (Clark et al., 1973; Prescott et al., 1971; Proudfoot and Wright, 1970).

1. Analytical Methods of Determining Paracetamol Concentrations

Paracetamol in biological fluids can be estimated by colorimetric (Brodie and Axelrod, 1948; Welch and Conney, 1965), spectrophotometric, gas-liquid, thin layer, and high performance liquid chromatographic methods (Adriaenssens and Prescott, 1978; Andrews et al., 1976; Cummings et al., 1967; Howie et al., 1977; Wiener, 1978).

It is important to avoid simple methods employing acid hydrolysis of the drug to p-aminophenol without prior solvent extraction, since this may give rise to gross overestimates in the presence of the acid-labile conjugates of paracetamol (Stewart et al., 1979). The chromatographic methods are preferable on the grounds of specificity, and simple, rapid and specific methods for the estimation of paracetamol and its sulphate, glucuronide, mercapturic acid and cysteine conjugates in plasma and urine are now available using high performance liquid chromatography (Adriaenssens and Prescott, 1978; Howie et al., 1977).

The pharmacokinetics of paracetamol in healthy subjects have been reported following intravenous and oral administration (Briant et al., 1976; Clements and Prescott, 1976; Clements et al., 1978; Cummings et al., 1967; Levy and Yamada, 1971; Lowenthal et al., 1976; Nelson and Morioka, 1963; Øie et al., 1975; Perucca and Richens, 1979; Prescott, 1980; Prescott and Wright, 1973; Rawlins et al., 1977; Slattery and Levy, 1979). A summary of some reported pharmacokinetic values is given in table I. These data are discussed in further detail in the text which follows.

2. Absorption of Paracetamol

2.1 Oral Absorption

Although paracetamol is rapidly absorbed from the gastrointestinal tract, it is incompletely available to the systemic circulation after oral administration, a variable proportion being lost through first-pass metabolism (Chiou, 1975; Perucca and Richens, 1979; Rawlins et al., 1977). The proportion of the dose reaching the systemic circulation appears to depend upon the amount administered, decreasing from 90% after 1 to 2g to 68% after 0.5g (Rawlins et al., 1977).

Paracetamol absorption depends on the rate of gastric emptying (Heading et al., 1973). A recently described pharmacokinetic model allows for the influence on paracetamol absorption of the various rates and patterns of gastric emptying (fig. 1). With this model there is excellent agreement between the observed and calculated plasma paracetamol concentrations with different rates and patterns of gastric emptying, and the analysis confirms that gastric emptying is rate limiting. The mean half-time for absorption from the small intestine is about 7 minutes, whereas gastric emptying half-times are about 3 times longer (Clements et al., 1978).

Although no perfusion studies have been carried out in man, paracetamol absorption appears to be negligible from the stomach but very rapid from the small intestine. In rats, absorption does occur slowly from the stomach and colon, and is most rapid from the small intestine, with 70% of the drug being absorbed within 30 minutes (Bagnall et al., 1979). Absorption is by passive transport with first-order kinetics and there appears to be no particular site of preferential absorption in the small intestine. Some metabolism may occur during absorption; Josting et al. (1976), for example, noted that 8% of the drug was metabolised by the perfused rat intestinal loop during a 30-minute period.

Mean maximal plasma paracetamol concentrations in fasting healthy subjects have been noted at 22 minutes after ingestion of paracetamol in solution (Nimmo et al., 1975a); and at 20 minutes (McGilveray and Mattack, 1972), 60 minutes (Dordoni et al., 1973) and 1.4 hours (Heading et al., 1973) after ingestion of paracetamol tablets. Although tablet formulation may influence the rate of paracetamol absorption (Gwilt et al., 1963b; Rawlins, 1981), this has not been confirmed in all studies (McGilveray et

Table I. Pharmacokinetic values for paracetamol in healthy adults[1]

Dose	No. of subjects	$t_{1/2\alpha}$ (min)	$t_{1/2\beta}$ (min)	V_1 (L/kg)	Vd_β	Clearance	f (%)[2]	Reference
Intravenous administration								
12mg/kg (injection)	4	7.8	119	0.51	0.90L/kg	5.45ml/min/kg	76	Clements and Prescott (1976); Prescott (1980)
1000mg (injection)	6	—	151	0.94	0.96L/kg	322ml/min	—	Perucca and Richens (1979)
1000mg (infusion; 5 min)	6	16	148	0.60	75.4L	352ml/min	—	Rawlins et al. (1977)
Oral administration								
500mg	6	—	167	—	—	534ml/min	63	Rawlins et al. (1977)
1000mg	28	—	105	0.86	—	5.7ml/min/kg	—	Briant et al. (1976)
1000mg	6	—	—	—	—	337ml/min	89	Perucca and Richens (1979)
1000mg	6	—	161	—	—	337ml/min	89	Rawlins et al. (1977)
1000mg/1.73m^2	3	—	142	—	86.0L/1.73m^2	400ml/min/1.73m^2	—	Lowenthal et al. (1976)
2000mg	6	—	139	—	—	331ml/min	87	Rawlins et al. (1977)

1 Volumes and clearances are not corrected for fraction of dose absorbed.
2 f (%) = fraction of dose reaching systemic circulation.

al., 1971; Richter and Smith, 1974). Under practical clinical conditions there may be as much as an 80-fold range in plasma concentrations 1 hour after ingestion of a therapeutic dose of tablets (Prescott, 1974). In addition, the rate of absorption may vary from one occasion to another; Richter and Smith (1974), for example, noted a 50% change in the 30-minute plasma paracetamol concentration when subjects were given the same dose on 2 occasions.

As paracetamol absorption seems to be predominantly dependent on the rate of gastric emptying, any drug, disease, or other condition which alters the rate of gastric emptying will influence the rate of paracetamol absorption (Nimmo, 1976).

2.1.1 Effect of Age, Weight and Sex

Although maximum plasma paracetamol concentrations have been noted to be lower in heavier subjects and highest in those aged 40 to 50 years (Gwilt et al., 1963b), these findings may merely be related to the fixed dose given to all subjects. After oral administration the maximal plasma paracetamol concentrations were found to be significantly higher in women in both the luteal and follicular phases of their ovulatory cycles than in men (Wójcicki et al., 1979).

2.1.2 Effect of Food

Increasing amounts of carbohydrate in the gut appear to delay paracetamol absorption (Jaffe et al., 1971; McGilveray and Mattock, 1972). Thus, in the same subjects, maximal plasma concentrations in the fasting state occurred 20 minutes after paracetamol ingestion, but were delayed to a mean of 90 minutes after a high carbohydrate meal at which time the mean peak plasma concentrations were approximately 25% of those observed under fasting conditions (McGilveray and Mattock, 1972). High lipid or high protein meals appear to have no such effect (Jaffe et al., 1971). Food does not, however, affect the total amount of the drug absorbed.

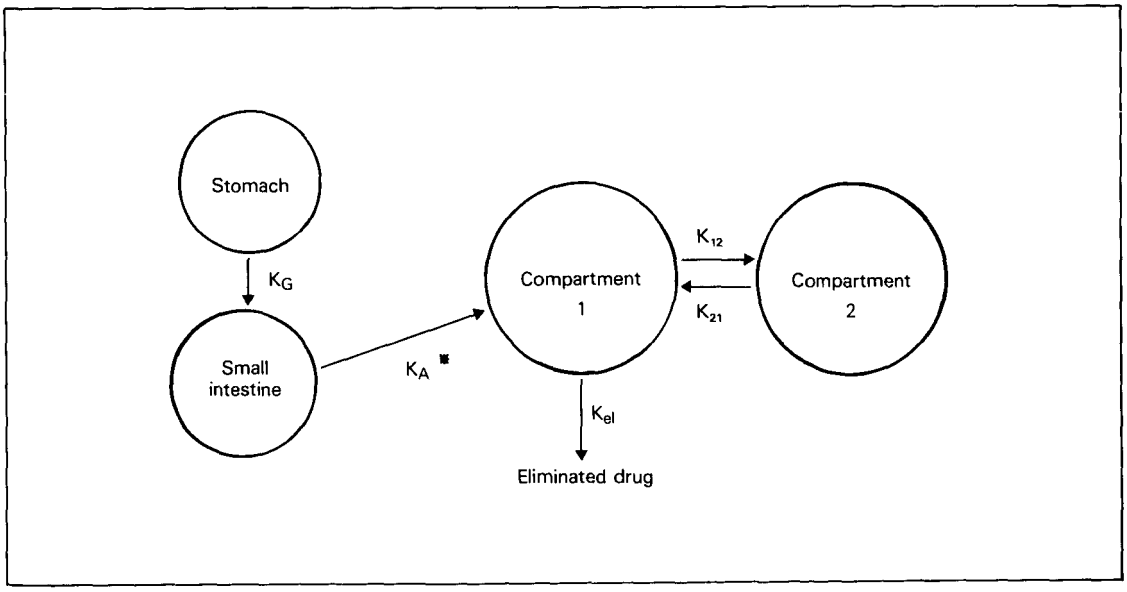

Fig. 1. Model for paracetamol absorption with 2 gut compartments.

First-order rate constants are: gastric emptying (K_G), intestinal absorption (K_A*), intercompartmental transfers (K_{12} and K_{21}) and elimination (K_{el}). Compartments 1 and 2 are the central and peripheral body compartments, respectively (after Clements et al., 1978).

2.1.3 Effect of Posture

Paracetamol absorption seems to be less rapid during sleep (McGilveray and Mattock, 1972). Nimmo and Prescott (1978) noted that the rate of paracetamol absorption was markedly reduced when subjects lay on their left side: at 15 and 30 minutes, respectively, paracetamol concentrations were 0.18 and 7.8μg/ml when subjects lay on their left side, compared with 12.5 and 20.8μg/ml, respectively, when ambulant.

2.1.4 Effect of Drugs

In healthy volunteers propantheline 30mg administered intravenously delayed the mean time to reach peak plasma paracetamol concentrations from 70 to 160 minutes whilst the mean maximum concentration was reduced from 26.3 to 17.5μg/ml. After intravenous administration of metoclopramide 10mg the mean time to reach the maximum plasma concentration was reduced from 120 to 48 minutes, while the mean maximum concentration increased from 12.5 to 20.5μg/ml (Nimmo et al., 1973). Total urinary recovery, however, was not affected by either drug.

Although paracetamol absorption is normal in early labour, it is markedly delayed when such patients are given pethidine, diamorphine or pentazocine; this delay is not reversed by intramuscular administration of metoclopramide (Nimmo et al., 1975b). In healthy volunteers, intramuscularly administered pethidine (150mg) and diamorphine (10mg) markedly delayed the absorption of paracetamol. The mean peak plasma concentration was 20μg/ml in the controls, 13.8μg/ml after pethidine and only 5.2μg/ml after diamorphine, whilst the mean times taken to achieve peak plasma concentrations were 22, 114 and 142 minutes, respectively. The total amount of paracetamol absorbed was not affected by either drug (Nimmo et al., 1975a).

Activated charcoal (10g) taken at the same time as paracetamol caused a more than 4-fold reduction in the mean maximum paracetamol concentration, and an average reduction of 63% in the area under the concentration-time curve for the period 0 to 120 minutes. There was a concomitant increase in the time taken to reach the mean peak plasma concentration, from 60 to 120 minutes. When charcoal was taken 60 minutes after paracetamol, the mean reduction in absorption was only 23%. Cholestyramine (12g) taken with paracetamol caused a similar reduction in the peak plasma paracetamol concentrations, with a mean reduction in absorption of 62%. This reduction was less marked (only 16%) when the drug was given 60 minutes after the paracetamol (Dordoni et al., 1973). Similar findings have been reported by Levy and Houston (1976), who noted that following the administration of 1g of paracetamol the total amount absorbed was reduced by 68% after 10g of charcoal and by 50% after 5g. A reduction in the rate of paracetamol absorption following the simultaneous ingestion of guar gum and pectin was probably due to a reduction in the rate of gastric emptying; total absorption was not reduced (Holt et al., 1979a). Codeine phosphate (15 to 32mg) does not affect the rate of paracetamol absorption (Bajorek et al., 1978; Nimmo et al., 1979).

2.1.5 Effect of Disease

Paracetamol absorption has been studied in only a small number of disease states. It is grossly impaired in pyloric stenosis (Prescott, 1974), but normal in patients with achlorhydria (Pottage et al., 1974). In patients with coeliac disease and Crohn's disease absorption was slower than in healthy subjects, largely due to slower gastric emptying (Holt et al., 1979b). In untreated thyrotoxicosis absorption is significantly faster than when the patients are euthyroid; absorption is only slightly delayed in hypothyroid patients (Forfar et al., 1980).

2.2 Rectal Absorption

Although paracetamol is well absorbed from the rectum, the rate of absorption is slower than following oral administration (Moolenaar et al., 1979a,b). With aqueous suspensions given rectally, the rate of absorption increased with increasing volume of the

enema; the pH of the enema fluid was not an important factor (Moolenaar, 1979a). When the enema volume was either 5 or 20ml, the peak plasma concentrations were 40 and 80%, respectively, of those following oral administration of the same dose (1000mg). Bioavailability was the same following oral and rectal administration. Absorption from paracetamol suppositories was found to be slower than from an enema; the smaller the particle size and the larger the suppository volume, the more rapid the absorption (Moolenaar et al., 1979b). Absorption from certain formulations of paracetamol suppositories is significantly slower when the suppository is 1 year old compared with when new (Moës and Jaminet, 1976).

3. Distribution and Protein Binding of Paracetamol

Paracetamol distributes throughout most tissues and fluids, reaching a tissue : plasma concentration ratio of about unity in all tissues except fat and cerebrospinal fluid (Brodie and Axelrod, 1949; Gwilt et al., 1963a). After intravenous administration, plasma paracetamol concentration-time curves are multiexponential and the relatively short half-times for the initial disposition phase ($t_{1/2\alpha}$ equal to 3 to 19 minutes) indicate rapid tissue distribution (Clements and Prescott, 1976; Rawlins et al., 1977). The apparent volume of distribution increases from 0.5L/kg bodyweight immediately after intravenous injection to 0.9L/kg after distribution is complete (Clements and Prescott, 1976; Perucca and Richens, 1979; Rawlins et al., 1977; table I). The distribution volume is similar in healthy subjects, in the elderly (Briant et al., 1976), in patients with epilepsy (Perucca and Richens, 1979), Gilbert's syndrome (Douglas et al., 1978) and in anephric patients (Lowenthal et al., 1976). In patients studied when thyrotoxic, hypothyroid or euthyroid, there were no significant differences between the groups with regard to distribution rate constants or apparent volume of the peripheral compartment, although the overall apparent volume of distribution tended to be largest in the thyrotoxic group and smallest in the hypothyroid group (Forfar et al., 1980). Following a therapeutic dose the concentration of paracetamol in whole blood has been reported to be 0 to 20% higher than that in plasma (Gazzard et al., 1973; Gwilt et al., 1963a; Perucca and Richens, 1979). Paracetamol apparently does not bind to plasma proteins at therapeutic plasma concentrations, but 15 to 21% is bound at plasma concentrations associated with overdosage (Gazzard et al., 1973). The sulphate and glucuronide conjugates do not bind to plasma proteins even at the relatively high concentrations found in anephric patients (Lowenthal et al., 1976). These 2 metabolites have a considerably smaller apparent volume of distribution than paracetamol (Lowenthal et al., 1976).

4. Elimination

Paracetamol is extensively metabolised (predominantly in the liver) with only 2 to 5% of a therapeutic dose being excreted unchanged in the urine. Although the drug is metabolised to a mintor extent in the rat gut (Josting et al., 1976) and kidney (Jones et al., 1979; Mitchell et al., 1977), it is not known whether this also occurs in man. The major metabolites are the sulphate and glucuronide conjugates, although a minor fraction is converted by cytochrome P-450-dependent hepatic mixed function oxidase to a highly reactive alkylating metabolite, which is probably N-acetyl-p-benzo-quinoneimine (Miner and Kissenger, 1979). This metabolite is usually rapidly inactivated by conjugation with reduced glutathione, and is eventually excreted in the urine as cysteine and mercapturic acid conjugates. Large doses of paracetamol (overdoses) cause acute hepatic necrosis as a result of depletion of glutathione and of covalent binding of the excess reactive metabolite to vital cell constituents (Mitchell et al., 1973, 1974).

In young healthy subjects approximately 85 to 95% of a therapeutic dose is excreted in the urine within 24 hours, with about 4, 55, 30, 4 and 4% appearing as unchanged paracetamol, and its

glucuronide, sulphate, cysteine and mercapturic acid conjugates, respectively (Cummings et al., 1967; Forrest et al., 1979; Jagenburg et al., 1968; Levy and Yamada, 1971; McGilveray et al., 1971; Mitchell et al., 1973; Prescott, 1980).

Other minor metabolites have been described, each accounting for 1% or less of a therapeutic dose. These include sulphate and glucuronide conjugates of 3-methoxy- and 3-hydroxy-paracetamol (Andrews et al., 1976; Mrochek et al., 1974) and 3-methylthioparacetamol (Klutch et al., 1978). Figure 2 shows the time relationships between the plasma concentrations of paracetamol and its sulphate and glucuronide conjugates in healthy subjects following an oral dose of 20mg of paracetamol per kg bodyweight. Sulphate and glucuronide conjugates are present in the plasma of all subjects at 15 minutes, but the peak concentrations of sulphate occurred earlier (1.4 ± 0.4 hours) than the glucuronide (2.4 ± 0.8 hours). The sulphate concentrations are lower than the paracetamol concentrations, whilst the glucuronide concentrations are higher.

Mean terminal elimination phase half-life values for paracetamol are in the range of 1.9 to 2.5 hours after intravenous administration of doses of 800 to 1000mg (table I), but the value found by non-linear regression analysis is dependent upon the method of weighting of individual data points (Clements and Prescott, 1976). Rawlins et al., (1977) noted no difference in the plasma half-life after different oral doses nor between intravenous and oral dosing.

The total body clearance values are about 5ml/kg/min (table I).

The kinetics of unchanged paracetamol can be determined from samples of saliva, since levels are similar to those in plasma (Glynn and Bastian, 1973; Lowenthal et al., 1976). However, glucuronide and sulphate conjugates have not been detected in saliva.

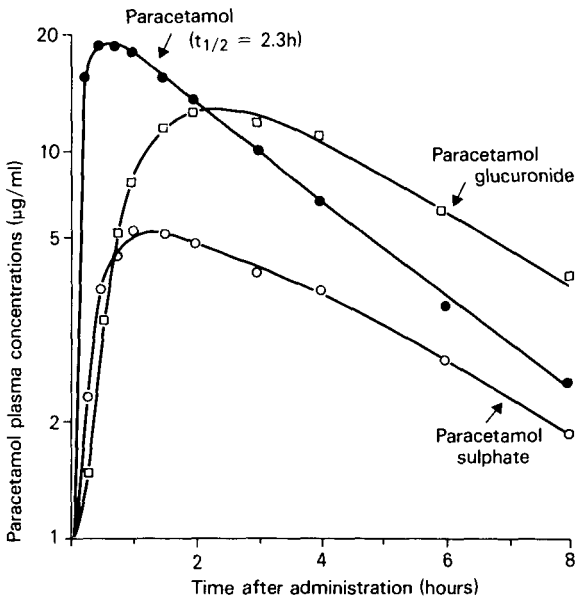

Fig. 2. Mean plasma concentrations of paracetamol and its sulphate and glucuronide conjugates (expressed as paracetamol equivalents) in 8 healthy subjects after an oral dose of 20mg/kg (after Prescott, 1980).

4.1 Effect of Age

Although the plasma paracetamol half-life, based on urinary excretion patterns, has been reported to be significantly prolonged in the newborn (Levy et al., 1975), this was not confirmed in a subsequent study (Miller et al., 1976). No significant differences were noted in the rate of paracetamol elimination between groups of children up to 12 years of age and adults (Alam et al., 1977; Miller et al., 1976). In all 3 studies, however, neonates and children aged 3 to 10 years excreted significantly less glucuronide and more sulphate conjugate than children aged 12 years and adults. In the children aged 10 years or less the rate constant for paracetamol glucuronide formation was smaller than in adults, whilst that for paracetamol sulphate formation was larger. The total excretion of paracetamol and its metabolites was the same at all ages.

In 2 studies the plasma paracetamol half-life has been noted to be significantly prolonged in geriatric subjects compared with younger subjects: 130 ± 9 minutes *versus* 109 ± 8 minutes (Briant et al., 1976) and 2.17 hours *versus* 1.75 hours (Triggs et al., 1975). The plasma half-lives in the geriatric subjects, however, are in the same range as those reported by other workers for healthy young subjects (Forrest et al., 1979; Prescott al., 1971; Rawlins et al., 1977). Although Briant et al. (1976) found the plasma clearance to be lower in geriatric subjects, this was not confirmed by Triggs et al., (1975).

4.2 Effects of Sex, Environmental or Genetic Factors and Drugs

The plasma paracetamol half-life has been noted to be slightly (15%), but significantly, longer in healthy males at 6am than at 2pm (Shively and Vessell, 1975). As the plasma clearance was not significantly different at these times, the difference in plasma half-life was presumably due to a change in the volume of distribution. The paracetamol half-life was found to be slightly, although not significantly, longer in women in the follicular phase (15%) and in the luteal phase (33%) of their ovulatory cycles, than in men (Wójcicki et al., 1979).

In a study of the effect of genetic and environmental factors on paracetamol metabolism, Mucklow et al. (1980) found the plasma half-life to be significantly longer, and the plasma clearance significantly slower, in Asians in London compared with Caucasian subjects. When considering all subjects the plasma clearance was significantly lower in men than in women, and it increased significantly with increasing alcohol intake and cigarette consumption, and was greatest in those women using the contraceptive pill. The effects of alcohol and the oral contraceptive pill were found in Caucasian subjects, but not in Asians. The plasma clearance was significantly faster in those who ate meat regularly than in vegetarians. Further analysis revealed only height as an independent correlate in the Asians. The variables which were found to correlate with paracetamol clearance accounted for only 27% of the total sample variance.

Mitchell et al. (1974) noted in healthy volunteers that pretreatment with phenobarbitone (phenobarbital) for 5 days had no effect on the plasma half-life of paracetamol. Although Perucca and Richens (1979) found the mean plasma half-life to be shorter in 6 epileptics (1.88 hours) than in 6 healthy subjects (2.52 hours), the difference was not significant. Prescott et al. (1981), however, found 15 'induced' patients (including 13 epileptics) to have a significantly shorter half-life (1.9 hours) than 12 healthy subjects (2.4 hours). In the epileptic subjects glucuronide conjugation was enhanced, but there was no significant difference in sulphate conjugation. The urinary excretion of the mercapturic acid and cysteine conjugates was the same in both groups, so that conversion of paracetamol to its potentially hepatotoxic metabolite did not seem to be increased in these patients with induced hepatic enzymes. Thus, although the hepatotoxicity of paracetamol is increased in most laboratory animal species by pretreatment with microsomal enzyme inducers such as phenobarbitone, 3-methylcholanthrene and ethanol (Jollow et al., 1974; Mitchell et al., 1973, 1974;

Streeter and Timbrell, 1979; Strubelt et al., 1978; Teschke et al., 1979), the results of Prescott et al. (1981) suggest that patients with induced hepatic enzymes are not at increased risk of hepatotoxicity from this cause when taking therapeutic doses of paracetamol. Although liver damage has been reported following the therapeutic use of paracetamol in chronic alcoholics (Gerber et al., 1980; Goldfinger et al., 1978; Licht et al., 1980; McClain et al., 1980), in many of these cases the drug had been taken in excessive doses. There is some evidence, however, that in paracetamol overdosage 'induced' patients may be at an increased risk of developing liver damage (Wright and Prescott, 1973).

4.3 Effect of Disease

4.3.1 Renal Disease

Plasma paracetamol concentrations and half-life values are similar in patients with chronic renal disease and in healthy subjects (Lowenthal et al., 1976; Øie et al., 1975; Prescott, 1969). The sulphate and glucuronide conjugates, however, accumulate in the plasma of patients with renal disease and reach concentrations 4 times those seen in healthy subjects (Lowenthal et al., 1976), but are readily removed by dialysis (Øie et al., 1975).

4.3.2 Thyroid Disease

In patients with untreated thyrotoxicosis, the mean plasma half-life was shorter, and the total body clearance of the drug was increased, compared with values obtained when the patients were subsequently euthyroid. In hypothyroid subjects there was only a slight reduction in the elimination rate (Forfar et al., 1980).

4.3.3 Liver Disease

Gilbert's disease: Although the plasma paracetamol clearance was found to be significantly lower in patients with Gilbert's disease than in healthy subjects, the plasma half-life was similar in the 2 groups. The plasma paracetamol concentrations declined monoexponentially in the patients and biexponentially in the controls (Douglas et al., 1978).

Acute liver disease: The plasma paracetamol half-life is grossly prolonged in patients with paracetamol-induced acute hepatic necrosis (see section 7).

Chronic liver disease: The elimination of paracetamol from the plasma in patients with chronic liver disease has been studied by a number of workers (Andreasen and Hutters, 1979; Arnman and Olsson, 1978; Fevery and de Groote, 1969; Forrest et al., 1977, 1979; Shamszad et al., 1975). These studies have shown that in patients with cirrhosis who have a normal plasma albumin concentration and prothrombin time, the plasma paracetamol half-life or clearance is similar to that seen in healthy subjects. Elimination is grossly abnormal, however, in patients with cirrhosis who have a low plasma albumin and an increased prothrombin time ratio. Thus, Forrest et al. (1979) found the mean plasma half-life to be 2.43 ± 0.19 hours in healthy subjects, 2.16 ± 0.54 hours in patients with chronic liver disease and a normal plasma albumin and/or prothrombin time ratio, and 4.25 ± 1.15 hours in patients in whom both these indices were abnormal. Good correlations have been demonstrated between the plasma half-life or clearance of paracetamol and the serum albumin concentration, prothrombin time ratio, plasma antipyrine (phenazone) and lignocaine (lidocaine) half-lives, and galactose elimination (Andreasen and Hutters, 1979; Forrest et al., 1979). A greatly prolonged plasma paracetamol half-life with very high concentrations of unchanged paracetamol and low concentrations of the glucuronide and sulphate conjugates was noted in a patient with a porto-systemic shunt; this finding is consistent with shunting and greatly reduced first-pass metabolism (Forrest et al., 1979).

After a single dose of paracetamol the 24-hour urinary excretion of paracetamol and its conjugates was not significantly different in patients with decompensated liver disease or in normal subjects (Forrest et al., 1979). In further studies in patients with chronic liver disease given regular therapeutic doses of paracetamol (1g 8-hourly for 3 or 5 days), there

was no cumulation of the drug in the plasma, and the plasma half-life was similar at the beginning and at the end of the study (Andreasen and Hutters, 1979; Forrest, 1980). There would thus appear to be little risk of cumulation and toxicity in patients with decompensated liver disease given therapeutic doses of paracetamol for a few days. There is no information available as to whether this may occur if such patients were treated for longer periods or with higher doses.

5. *Urinary Excretion of Paracetamol and its Metabolites*

As paracetamol is a moderately lipid-soluble weak organic acid which is not extensively bound to plasma proteins, it is likely to undergo considerable glomerular filtration with subsequent passive tubular reabsorption. As predicted by the pK_a value of 9.5, the renal clearance of paracetamol is not dependent on urine pH, but there is a weak but significant correlation with flow rate. In healthy adults the renal clearance of paracetamol at normal urine flow rates is about 12ml/min (range 5 to 20ml/min). Forced diuresis is of no value in the treatment of paracetamol poisoning.

The highly polar glucuronide and sulphate conjugates appear to be both filtered at the glomerulus and actively secreted by the tubules since their renal clearances are approximately 130 and 170ml/min, respectively. The clearance of these metabolites is independent of changes in urine flow and pH (Prescott, 1980; Prescott and Wright, 1973).

6. *Biliary Excretion of Paracetamol*

Little is known of the biliary excretion of paracetamol and its metabolites in man. In rats, however, the glutathione conjugate is excreted into the bile, hydrolysed to the cysteine conjugate in the small intestine, and subsequently reabsorbed (Grafström et al., 1979).

7. *Paracetamol Overdose*

As there are no specific clinical manifestations of paracetamol poisoning, and maximum abnormalities of liver function tests are delayed for at least 3 days, the plasma paracetamol concentration is often used as a guide to prognosis. However, there is considerable individual variation in susceptibility to paracetamol hepatotoxicity. The probability of severe liver damage with plasma aspartate and alanine aminotransferase concentrations exceeding 1000iu/L is about 60% with plasma paracetamol concentrations above a 'treatment' line on a semilog graph joining plots of 20μg/ml at 4 hours and 30μg/ml at 15 hours after ingestion. In patients with plasma paracetamol concentrations above a parallel line joining plots of 300μg/ml at 4 hours at 45μg/ml at 15 hours, the risk of severe liver damage without specific treatment is 90% (Prescott et al., 1979). Liver damage following paracetamol overdosage can be prevented by early treatment (within 8-10h of the overdosage) with sulfhydryl compounds like N-acetylcysteine and methionine (Crome et al., 1976; Prescott et al., 1979).

Paracetamol metabolism is impaired from the outset in patients who develop liver damage after overdosage, and prolongation of the half-life is related to the severity of the hepatic injury. Some degree of hepatic necrosis is likely if the plasma paracetamol half-life exceeds 4 hours and fatal hepatic encephalopathy may occur if it is prolonged beyond 10 to 12 hours (Gazzard et al., 1977; Prescott and Wright, 1973; Prescott et al., 1971). In patients with very severe liver damage there may be a progressive increase in the plasma half-life. This impairment of paracetamol metabolism seems to reflect almost immediate interference with hepatocyte function rather than concentration-dependent saturation of glucuronide conjugation as proposed by Slattery and Levy (1979). Prolongation of the plasma paracetamol half-life is associated with a marked increase in the ratio of the plasma concentrations of unchanged to conjugated drug, and in patients with fatal hepatic necrosis the conjugation of paracetamol may virtually cease (Prescott and Wright, 1973).

Table II. Percentage recovery of paracetamol and its sulphate, glucuronide, cysteine and mercapturic acid conjugates in healthy subjects given an oral dose of 20mg/kg, and in paracetamol-poisoned patients receiving different treatments (after Prescott, 1980)

Treatment	No. of subjects or patients	Recovery (%) of				Mean recovery (g)
		paracetamol	sulphate	glucuronide	mercapturic plus cysteine	
Healthy subjects (20mg/kg)	8	5.0	32.3	54.7	8.0	1.27
Poisoned patients without liver damage						
Supportive	9	8.6	9.3	75.3	6.9	10.3
Early cysteamine	15	9.3	11.9	72.4	6.4	14.5
Early N-acetylcysteine	23	11.2	14.5	63.6	10.6	13.4
Poisoned patients with severe liver damage						
Supportive	8	8.7	9.7	66.5	15.2	11.8
Late N-acetylcysteine	8	9.4	13.3	60.9	16.5	13.0

In contrast to glucuronide formation, sulphate conjugation is saturated immediately after an overdosage of paracetamol. Plasma concentrations of paracetamol sulphate are of the same magnitude in poisoned patients as in healthy volunteers taking therapeutic doses. However, sulphate conjugation is increased by treatment with sulfhydryl compounds, which presumably provide inorganic sulphate.

The changes in paracetamol metabolism following overdosage are reflected in the pattern of urinary excretion of its metabolites (table II) [Prescott, 1980]. Patients receiving supportive therapy only, who do not suffer liver damage, excrete much less paracetamol sulphate and more glucuronide that healthy volunteers given a dose of 20mg/kg. There is no significant change in the fraction recovered as the mercapturic acid and cysteine conjugates which reflect conversion of paracetamol to its hepatotoxic metabolite. In poisoned patients with severe liver damage, however, mercapturic acid and cysteine conjugate excretion is increased at the expense of glucuronide conjugation.

Similar findings have been reported by Davis et al. (1976).

These patterns may be modified by treatment with sulfhydryl compounds. Early effective treatment with cysteamine tends to reduce the excretion of the mercapturic acid and cysteine conjugates, while N-acetylcysteine increases the formation of the sulphate, mercapturic acid and cysteine conjugates with a corresponding fall in glucuronide excretion (table II). Methionine had no such effects (Prescott, 1980). These findings are consistent with the observations that cysteamine and N-acetylcysteine (but not methionine) form adducts with the intermediate metabolite, and that cysteamine also inhibits the oxidative metabolism of paracetamol (Buckpitt et al., 1979).

8. Drug Interactions with Paracetamol

When paracetamol is given concomitantly with chloramphenicol, the plasma chloramphenicol half-life is prolonged 5-fold (Buchanan and Moodley, 1979).

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